

Mobile Engineering

Module 1: Android Hello World, Gradle, and Build Tools

- **Documentation:** Read the [Meet Android Studio Guide](#) and explore the official [Configure Your Build Documentation](#).
- **Hands-on:** Complete the interactive [Build Your First Android App Codelab](#).
- **YouTube Resource:** [Watch Philipp Lackner's Gradle Guide for Android Developers](#).

Module 2: Version Control and Team Collaboration

- **Documentation:** Master the basics via [GitHub's Git Documentation](#) and check out the Android-specific flow in Android Studio Version Control Integration.
- **Hands-on:** Practice branching and merging dynamically with the [Learn Git Branching Playground](#).
- **YouTube Resource:** [Watch Amigoscode's Git and GitHub Full Tutorial](#).

Module 3: UI Components, Layouts, Navigation, and Views

- **Documentation:** Review the traditional [Android View Layouts Guide](#) and the foundational [Application Fundamentals](#).
- **Hands-on:** Work through the [Views and Layouts Codelab Collection](#).
- **YouTube Resource:** Watch Stevdza-San's [Android Jetpack Navigation Component Tutorial](#).

Module 4: Architecture Patterns and State Management

- **Documentation:** Absorb the modern [Guide to App Architecture](#) along with UI Layer State Management.
 - **Hands-on:** Complete Google's deep-dive [Android Advanced Architecture Blueprint](#).
 - **YouTube Resource:** Watch Philipp Lackner's [MVI vs MVVM Architecture Guide](#).
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Module 5: Storage, Data Persistence, and Networking

- **Documentation:** Read about local databases via the [Room Persistence Library](#).

Guide and networking via Square's Official Retrofit Documentation.

- **Hands-on:** Code along with the **Persist Data with Room Codelab**.
- **YouTube Resource:** Watch CodingWithMitch's Retrofit with Clean Architecture Series

Module 6: Dependency Injection and Common Android Libraries

- **Documentation:** Review Google's recommended **Dependency Injection with Hilt Guide**.
- **Hands-on:** Walk step-by-step through the **Using Hilt in your Android App Codelab**.
- **YouTube Resource:** Watch Philipp Lackner's Hilt Dependency Injection from Scratch.

Module 7: Testing, Debugging, and Quality Gates

- **Documentation:** Study the comprehensive **Test Apps on Android Guide** and Android Studio's **Logcat Debugging Suite**.
- **Hands-on:** Implement robust tests following the **Android Testing Codelab**.
- **YouTube Resource:** Watch Mitchell Tabian's Unit Testing for Android Guide.

Module 8: Distribution, App Release, and Store Deployment

- **Documentation:** Follow the launch criteria in the **Publish Your App Guide** alongside the **Android Developer Verification Mandates**.
- **Hands-on:** Earn your badge via the Google Play Store Listing Certificate Pathway.
- **YouTube Resource:** Watch any walkthrough from the Official Google Play Console YouTube Channel.

Module 9: Jetpack Compose and Modern UI Development

- **Documentation:** Read the complete **Jetpack Compose Foundations** manual and keep track of updates using the **Jetpack Compose Roadmap**.
- **Hands-on:** Go through the comprehensive **Android Basics with Compose Course**.
- **YouTube Resource:** Watch CheezyCode's Jetpack Compose Full Playlist for Beginners.

Module 10: Platform Strategy and Advanced Mobile Engineering

- **Documentation:** Prepare for modern scaling using the **Kotlin Multiplatform (KMP) Documentation** and build contextual cross-device software via the Android Adaptive Layouts Guide.
- **Hands-on:** Get your hands dirty with the **Build a Kotlin Multiplatform App Tutorial**.
- **YouTube Resource:** Watch Kotlin Multiplatform